

Bernardo Pereira

Computer Science & Engineering Student

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I'm a Computer Science and Engineering student specializing in distributed systems and cybersecurity. I enjoy working on reliable software where performance, correctness, and real-world impact are tightly connected.

Education

Instituto Superior Técnico <i>MSc, Computer Science and Engineering</i> Current Grade: 17/20	09/2025 – 07/2027
Instituto Superior de Engenharia de Lisboa <i>Bsc, Computer Science and Engineering</i> Grade: 17/20	09/2022 – 07/2025

Professional Experience

Sky <i>Apprentice Developer – xTV Devices & Tech</i> - Developed a Deeplink Testing Tool using Lightning.js, enabling multi-device deeplink validation across Peacock, SkyShowtime, NOW, and Showmax; reduced manual testing time by ~90%. - Gained practical front-end experience, adhering to agile methodologies and version control best practices.	07/2025 – 08/2025
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Projects

HtmlFlow 🔗 <i>Java/Kotlin DSL for type-safe HTML</i> - Maintainer of HtmlFlow, a high-performance Java/Kotlin DSL for compile-time safe HTML 5.2 generation. - Implemented hot-reload for views, enabling live UI updates without server restarts via hot-swapping. - Designed the ViewFactory API and automatic view discovery loader to simplify framework integration. - Added http4k support and improved reactive/coroutine-based rendering. - Upgraded CI/CD and code quality tooling using GitHub Actions and SonarCloud.	06/2025 – Present
Dida Meetings 🔗 <i>Distributed meeting management system</i> - Implemented a fault-tolerant distributed meeting manager using Paxos and Vertical Paxos for dynamic reconfiguration; built server, client, and admin console communicating over gRPC/Protobuf. - Implemented Multi-Paxos per-log-slot with leader rotation schedules, enabling reconfiguration and strong ordering for open/add/close operations.	09/2025 – 10/2025
Non-Blocking Progressive Server-Side Rendering Benchmark 🔗 <i>Final Project – BSc in Computer Science and Engineering (20/20)</i> - Designed and executed benchmarks comparing Spring MVC, WebFlux, and Quarkus across blocking, reactive, coroutine, and virtual-thread execution models; measured latency, throughput, and resource usage. - Demonstrated virtual threads provide scalable non-blocking rendering with blocking template engines, achieving comparable throughput to other non-blocking approaches. - Produced reproducible benchmark suite and documentation used by peers for follow-up experiments.	01/2025 – 07/2025

Skills

Languages Kotlin, Java, Typescript, Javascript, Go.	Backend Spring MVC & WebFlux, RxJava, Quarkus, Express.js, Nginx.
Frontend HTML, CSS, React, Lightning.js Material-UI, Jetpack Compose, Android.	Databases & Cloud PostgreSQL, MongoDB, Elasticsearch, Firestore, Google Cloud.

Awards

Merit Scholarship Next Level@IPL - 2022/2023 & 2024/2025

Languages

Portuguese • English — C1